Module 1: Optimizing Conditions for Bacterial Growth Assignment

Objective and Design

The purpose of this experiment is to determine how five different factors affect the growth of bacteria, and to identify the best combination of factor levels that maximizes bacterial yield. The factors considered in this analysis are: shaking (yes or no), glucose concentration (low=1g/L, high=5g/L), temperature (low=30°C, high=37°C), incubation time (low=12 hours, high=24 hours), and pH level (low=6, high=7.5). The response variable is Growth. Shown below is the JMP design table showing the responses, factors, run order, and number of runs.

Σ	Pattern	Temperature (Celsius)	pH Level	Glucose (g/L)	Shaking	Incubation Time (Hours)	Growth
1	+++1−	37	7.5	5	No	12	0.84
2	+−+1−	37	6	5	No	12	0.8
3	−+−2+	30	7.5	1	Yes	24	0.85
4	−−−1+	30	6	1	No	24	0.59
5	++−2−	37	7.5	1	Yes	12	0.81
6	−+−1+	30	7.5	1	No	24	0.68
7	+−+1+	37	6	5	No	24	0.86
8	−−−1−	30	6	1	No	12	0.51
9	+++2+	37	7.5	5	Yes	24	1.17
10	−−−2−	30	6	1	Yes	12	0.66
11	+−+2−	37	6	5	Yes	12	0.92
12	−−−2+	30	6	1	Yes	24	0.81
13	+−−1−	37	6	1	No	12	0.58
14	+−−2−	37	6	1	Yes	12	0.73
15	−+−1−	30	7.5	1	No	12	0.57
16	++−1+	37	7.5	1	No	24	0.77
17	−−+1+	30	6	5	No	24	0.71
18	−++1+	30	7.5	5	No	24	0.75
19	+−−2+	37	6	1	Yes	24	0.85
20	−−+2−	30	6	5	Yes	12	0.8
21	+−−1+	37	6	1	No	24	0.7
22	−−+1−	30	6	5	No	12	0.62
23	+++1+	37	7.5	5	No	24	0.93
24	−++1−	30	7.5	5	No	12	0.7
25	+−+2+	37	6	5	Yes	24	1.02
26	++−2+	37	7.5	1	Yes	24	0.97
27	−++2+	30	7.5	5	Yes	24	0.97
28	−+−2−	30	7.5	1	Yes	12	0.72
29	++−1−	37	7.5	1	No	12	0.67
30	+++2−	37	7.5	5	Yes	12	0.99
31	−−+2+	30	6	5	Yes	24	0.92
32	−++2−	30	7.5	5	Yes	12	0.82

Model and Analysis

The Summary of Fit shows the R^2 = 0.992, which means the model explains 99.2% of the variation in bacterial growth. This indicates that this model fits the data well and explains the variation in observed bacterial growth. The adjusted R^2= 0.985, which takes the number of predictors in the model into account.

Summary of Fit

RSquare	0.992046
RSquare Adj	0.984588
Root Mean Square Error	0.018498
Mean of Response	0.790313
Observations (or Sum Wgts)	32

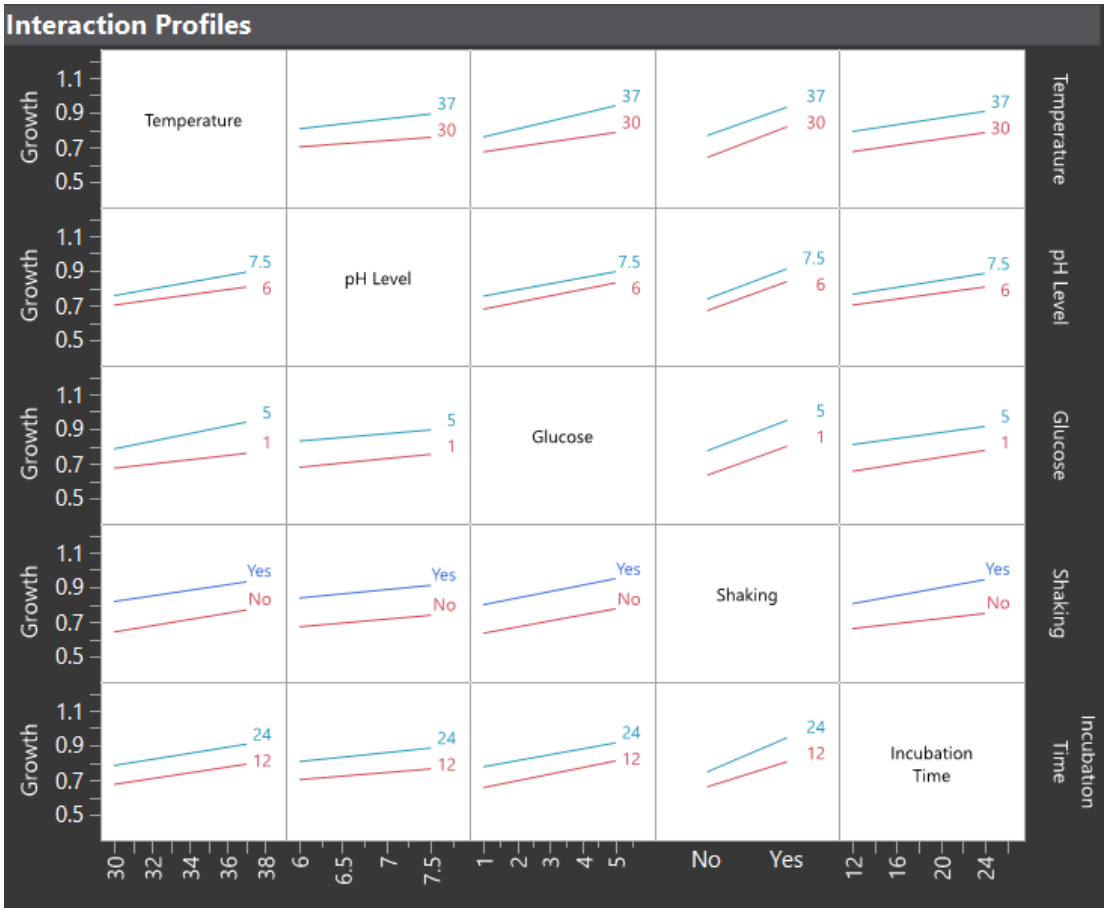
The Effect Summary displays the significance of each factor in the model, and the two way interactions of the factors. From it we see that the following factors are statistically significant in the growth of bacteria, in order of descending significance: shaking (LogWorth=13.813), glucose(LogWorth=12.798), temperature(LogWorth=11.478), incubation time(LogWorth=11.052), pH level (LogWorth=8.030), the interaction of temperature and glucose (LogWorth=4.106), the interaction of shaking and incubation time (LogWorth=2.911), and the interaction of temperature and pH level(LogWorth=1.529). Each of these factors and interactions had a P-Value < 0.05, indicating statistical significance. The remaining 7 interactions have P-values greater than 0.05.

Effect Summary

Source	Logworth	PValue
Shaking	13.813	0.00000
Glucose(1,5)	12.798	0.00000
Temperature(30,37)	11.478	0.00000
Incubation Time(12,24)	11.052	0.00000
pH Level(6,7.5)	8.030	0.00000
Temperature*Glucose	4.106	0.00008
Shaking*Incubation Time	2.911	0.00123
Temperature*pH Level	1.529	0.02955
pH Level*Incubation Time	0.634	0.23202
Glucose*Incubation Time	0.634	0.23202
Temperature*Shaking	0.510	0.30878
pH Level*Glucose	0.395	0.40245
Glucose*Shaking	0.290	0.51307
Temperature*Incubation Time	0.194	0.63924
pH Level*Shaking	0.194	0.63924

Results and Findings

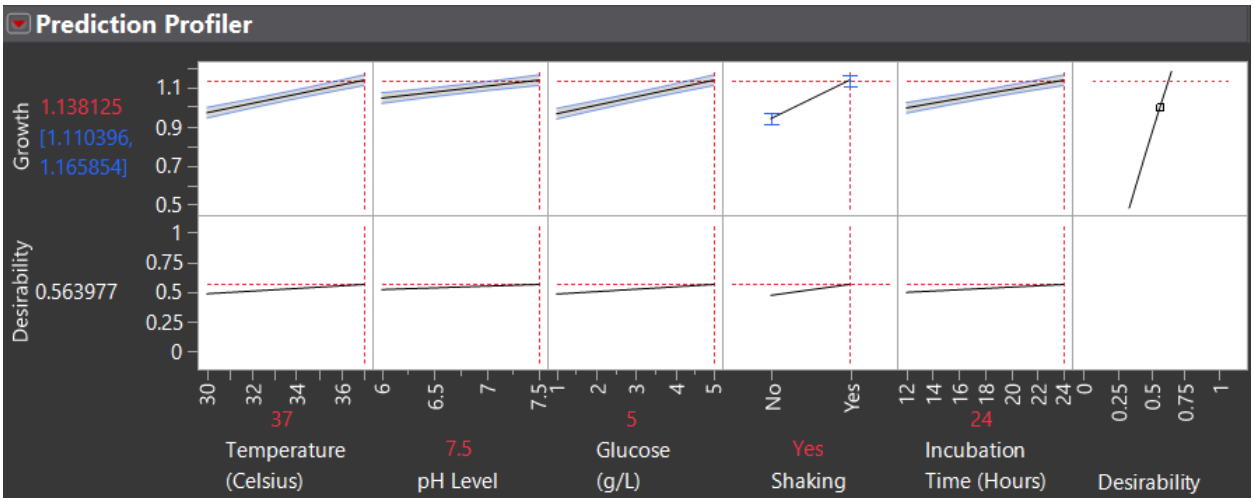
Based on the Effect Summary we know the interactions temperature and glucose, shaking and incubation time, and temperature and pH are statistically significant. The Interaction Profiles show how the effect of one factor on the response changes depending on the level of another factor, and can be determined by the slope of the lines. If the lines in a given pair are parallel there is no interaction between the two factors. The more different the slope of the lines are the more strong the interaction between the two factors. Seen below are the Interaction Profiles of all factors in the analysis. The blue lines represent the high value for each factor, the red lines represent the low value. The temperature and glucose interaction profile shows the blue line with a larger slope than than the red, showing that a positive interaction exists between temperature and glucose. Shaking and incubation time also shows faster growth when shaking and a 24 hour incubation time than shaking and only a 12 hour incubation time. In other words, once again the blue line has a larger slope than the red. Temperature and pH are similar. However, even though the blue line (pH=7.5) has a larger slope, the two pH values are closer together than other factors in each interaction profile. Though the lines are close together, the growth rate is much higher as temperature



increases when the pH=7.5 than when the pH=6. Most other interactions show parallel lines, indicating no relationship between the factors.

Prediction

As shown in the Prediction Profiler below, the following combination of factor levels predicts the highest growth: temperature=37°C, pH=7.5, glucose=5g/L, shaking=yes, and incubation time=24 hours. The maximum predicted growth value is 1.138, with a confidence interval of 1.11-1.17 (rounded).



Conclusion

The growth of bacteria is influenced by shaking, glucose levels, temperature, incubation time, and pH, with shaking and glucose having the largest effects. Statistically significant interactions were observed between temperature and glucose, shaking and incubation time, and temperature and pH. The effect of one factor depends on the other, and that when combined, the factors in these pairs increase the growth. According to the Prediction Profiler, the optimal conditions for maximum bacterial growth are 37°C, pH 7.5, 5 g/L glucose, shaking, and 24-hour incubation. Under these conditions, the predicted Growth reaches 1.138.